

# XAI-AMD-DL: An Explainable AI approach for Android Malware Detection System using Deep Learning

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**Abstract**— Efficient malware identification is essential to safe the system resources and privacy of data for cybersecurity system. The use of android smartphones has increased tremendously that attracting various types of malware attacks. Nowadays, malware writers use Artificial Intelligence (AI)-enabled malware attack techniques to bypass the detection of malicious activities. Hence, designing an efficient, effective and robust malware detection system to identify malware variants remains a critical problem and challenge. However, number of deep learning (DL) models applied for effective android malware detection in existing methods at large scale, but these models actually lacks interpretability of the models to explain the contribution of each features to the detection system. Therefore, this paper propose an Explainable Artificial Intelligence (XAI) based hybrid Convolutional Neural network (CNN) and Bi-Gated Recurrent Unit (Bi-GRU) Android Malware Detection (AMD) System using DL models named as XAI-AMD-DL. The proposed model is evaluated the using the CICAndMal2019 android malware dataset. The results obtained by the proposed XAI-AMD-DL model is 97.98% accuracy, and 97.75 %, 97.76%, 97.75% precision, recall and f1score, respectively outperforms the existing DL models.

**Keywords**—Explainable AI, Android Malware Detection, Deep Learning, Bi-GRU, Security and Privacy, CNN

## I. INTRODUCTION

A malware is software codes of malicious intension about the system resources, files, information, and password to theft valuable data and information[1][2]. Malware writer uses evasion techniques to create the new variants of malware to bypass the detection model. There are two commonly used malware analysis method. The one is Static analysis that detect the malware without executing the codes. However, this approach is ineffective for new variants of malware.

Second method is dynamic analysis used for analysis of new variants of malware by executing codes in virtual environment[3].

In recent times, Android Operating System (AOS) are being used by large number of users on daily basis[4]. AOS is open source software platform for mobile devices supported by Google Company. The android system has huge number of mobile users due to its popularity and easy to use. The application of android consists of many library files and records. The information of AOS is contains by manifest.xml file such as the bundle name, permission activities, segment information like service utility of applications, content providers, broadcast receivers, and other files. The directory in AOS used for keeping the images, icons, user interface. For every android application before accessing the information of the devices it requires authorization by operating system to ensure transparency and security. Android malware detection performed using the analysis of features such API calls, permission and intents by monitoring the behaviour of application software activity like malware or benign[5]. The major contribution of this paper are as below:

- Proposed an XAI-enabled android malware detection framework by leveraging DL based hybrid CNN and Bi-GRU model for cybersecurity.
- A new proposed XAI approach is applied for explaining individual feature contribution and brings the transparency in the overall results of the model.
- The experimental results analysis are performed with existing DL methods of malware detection and identification using CICAndMal2019 android malware datasets.

The other section of the paper consist of related work in section 2, proposed work in section 3, experimental work and results analysis in Section 4, and conclusion in Section 5.

## II. RELATED WORK

In paper [6] author proposed a light weight android malware detection framework. It select the vulnerable feature using non-negative matrix factorization method to obtained 50 lower dimensional feature among the 209 feature set. The mapping techniques mapped each API calls to specific feature for static analysis. Then, all this features were aggregated to count the frequency of each feature. In paper [7] author presented a machine learning(ML) and bio inspired optimization based android malware detection model. The proposed model applied the SVM and Artificial Bee Colony (ABC) optimization algorithm.

The author in paper [8] proposed a new method known as artificial malware based detection to handle the detection of obfuscation techniques used by attackers. The proposed method used the extracted feature and artificially malicious patterns generated by evolutionary genetic algorithm. By using original features and artificial malicious features, the proposed model can be detected new variants of malware. In [9] proposed an advanced hybrid malware detection model based on voting rule to build malware decision model. The hybrid model is combination of random forest algorithm and DL model for classification as benign or malicious files. The author in [10] proposed the enhanced function call graph method to monitor the run time behavior of the application. This work developed the graph convolutional based algorithm to obtain the feature vector representation of function call graphs and improved the performance in detection of malicious application.

The paper [11] addressed the concern of new malware type HummingBad that is capable to target multiple attacks for android platform silently. Therefore, proposed a ML based model to classify malware by using opcodes features extracted from application. In [12] author worked on feature weight mapping and joint optimization of selected feature which is essential to enhanced the performance of AMD model. The presented framework calculates the weight of each feature of the application, and then used joint parameter optimization method that is differential evolution algorithm to improve the accuracy of model. Another work in [13] presented two approaches for static analysis of android malware app detection using classification and clustering ML techniques. These two approaches worked on two

different feature of android mobile device architecture , first approaches uses the permission feature to allow access into android system for further operation whether app is malicious or benign. Another approach scans the source code by using the bag-of words method.

## III. PROPOSED METHOD

This section explains the proposed work, it consist of the three sub sections: a data pre-processing, description of hybrid CNN and Bi-GRU model and about XAI approach used for AMD. The Fig. 1 depicts the proposed XAI-AMD-DL framework. The presented framework aims to detect obfuscated and polymorphic malware variants. In this work a new XAI approach is used for interpretation of the features of the detection system, which show the importance of features in high to low order. This proposed work is based on static analysis that uses intents and permission as a features. The availability of big data increased the uses of DL models to detect malware effectively in recent days.

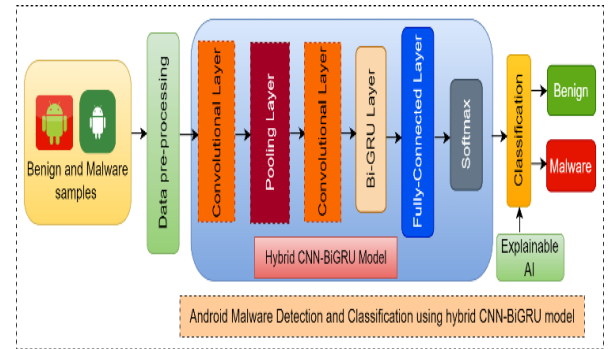


Fig. 1 Proposed AMD Model

### A. Pre-processing

It is to eliminate the meaningless values, simplify and transform data to numerical values. In this work, initially we have applied feature mapping to convert the categorical data into numeric value, replace the missing value with unknown class and by mean of given feature value[14]. Then, the min-max normalization is applied to rescale the data values between range of 0 and 1. The minimum values get changed into 0, the maximum value get changed into 1 and remaining feature values gets converted into decimal points between 0 and 1. Eq. (1) represents the mathematical formula[15][16].

$$A_{new} = \frac{A - A_{min}}{A_{max} - A_{min}} \quad (1)$$

Here, A is the feature values that need to be reduce,  $A_{max}$  representing the highest feature value and  $A_{min}$  is the lowest feature values in the dataset[17].

### B. Description of CNN

The CNN model[18] is consist of convolution layers, pooling layers, a fully connected layer. The task of feature extraction is performed by the convolutional and the pooling layers. The output layer is the fully connected layer which flatten all the neuron in single feature vector. CNN model[19] consist of two convolutional layers that made of set of filters which act upon the input data, activation function is used to obtain feature map vector and one pooling layer applied to decrease the dimension of feature map vector and computational complexity[20]. The best quality of the CNN model is to simplify the training phase, which is obtained by updating weights of both layers in multiple cycles [21].

### C. Description of Bi-GRU

The presented work employed Bi-GRU model for malware detection and identification. Bi-GRU is a gating mechanism used in recurrent neural network similar to LSTM model. The Bi-GRU having two gates such as reset gate and update gate used to modify the past and present state of the gates, and it requires less number of parameter as compared to LSTM model that keeps three gates (input, forget and output)[22]. It is capable to handle long-term dependency and multiple time sequence job at large scale. The update gate used to trade-off between the quantity of information of past and present state. A maximum value of the update gate pass more information from the last state to the present state. Whereas, the reset gate is responsible for fixing the quantity of last state information to be removed. A minimum value of the reset gate indicates that less state information is rejected. The mathematical expression of Bi-GRU can be represented in the Eq. (2)-Eq. (6). Where  $u_t$  represents update gate and  $r_t$  is reset gate.  $m_r, m_u, m_h$  and  $m_o$  are the weight matrix of the reset gate, update gate, hidden state and output state respectively, and  $d_u, d_r, d_h$  are the deviation vector[21].

$$r_t = \sigma(m_r * [h_{t-1}, p_t] + d_r) \quad (2)$$

$$u_t = \sigma(m_u * [h_{t-1}, p_t] + d_u) \quad (3)$$

$$\tilde{h}_t = \tanh(m_h * [r_t * h_{t-1}, d_h]) \quad (4)$$

$$h_t = (1 - u_t) * h_{t-1} + u_t * \tilde{h}_t \quad (5)$$

$$q_t = \sigma(m_o * h_t) \quad (6)$$

Here,  $p_t$  is the input at time  $t$  and  $h_{t-1}$  represents hidden state at time  $t - 1$  which consists of the last and next nodes information. The output is indicated by  $q_t$  at time  $t$  and the hidden state is represented by  $h_t$  move to the next node.  $\sigma$  indicate the sigmoid function that changes the input data in the range of 0 and 1,  $\tanh$  is the activation function that rescale the data between  $-1$  and  $1$ .

### D. Explainable AI

The Explainable AI termed as XAI or interpretability model, which is applied in this work to bring the transparency in the results of the model by the contribution of individual features[23]. It is a game theory based feature visualization approach called as Shapley Additive Explanations (SHAP). It is effective in describing the supervised learning models. The

proposed model incorporate the Kernel SHAP to describe features importance detected by the Bi-GRU model. The SHAP method explains accurate explanation of important features by establishing relationship among the features with maximum reconstruction error and the features that are most informative. The shapely values are generated by game theory to explain a particular prediction by allocating the SHAP value to individual feature that keeps the following three characteristics such as: local accuracy, missingness and consistency[24].

## IV. EXPERIMENTAL WORK

This section demonstrates the experiment and result analysis of the proposed XAI-AMD-DL model for malware detection. In experimental work used the python programming language, The proposed work used CICAndMal2019 android malware dataset[25] for evaluation purpose using the most preferred metrics[26] as represented in Table 1. The proposed model is trained with 70% train data and tested using 30 % test samples for binary classification. The proposed model is compiled with number of hyper parameters as shown in Table 2 such as ‘Adam’ optimizer to minimize the loss of the model, tanh activation function used to regulate the output between  $-1$  and  $1$ , softmax function used for classification of malware and benign samples.

The proposed hybrid CNN-BiGRU model further interpreted with SHAP framework for feature analysis. The results of applied SHAP framework for feature importance is in Fig. 2 depicted the dot plot of top ten important features in decreasing order (high to low), which shows the impact on model output and Fig. 3 indicated the bar plot of top twenty important features in decreasing order (high to low), that contributed the average impact on model output to the overall result. For example in Fig. 3 and Fig. 4 android.telephony.SmsManage show the highest impact in result of model output and second highest impactful feature is android.telephony.gsm.SmsManager. The performance of the proposed model are also mentioned in terms of confusion matrix to calculate precision and recall, model loss and model accuracy in Fig. 4 to Fig. 6 for comprehensive results analysis. The obtained results of proposed model is 97.98% accuracy, precision is 97.75%, recall is 97.76% and f1score is 97.75% which is better than the existing DL models as compared in Table 3. The obtained result demonstrates the presented model is effective for AMD.

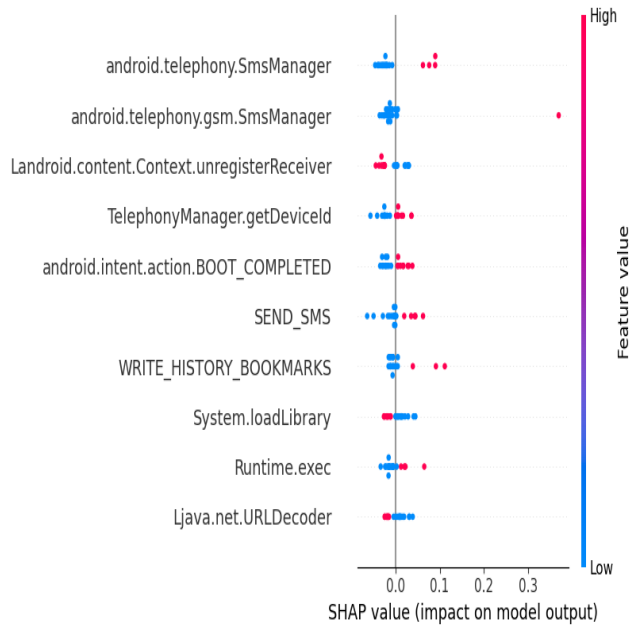


Fig. 2 Dot plot of Feature Importance

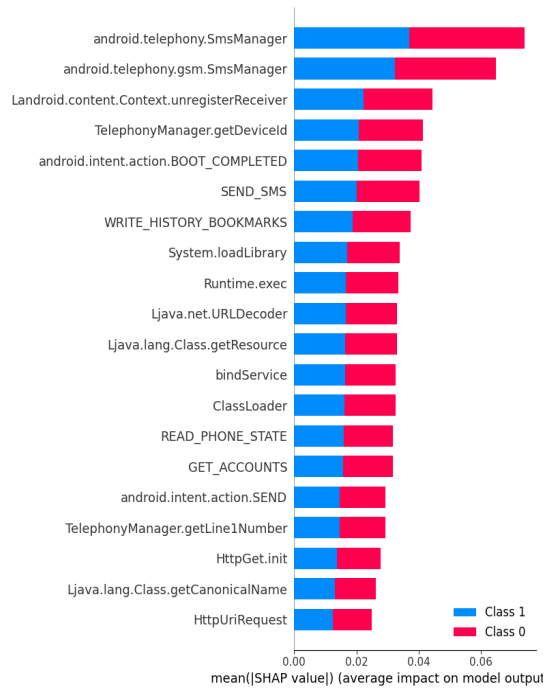


Fig. 3 bar plot of important features

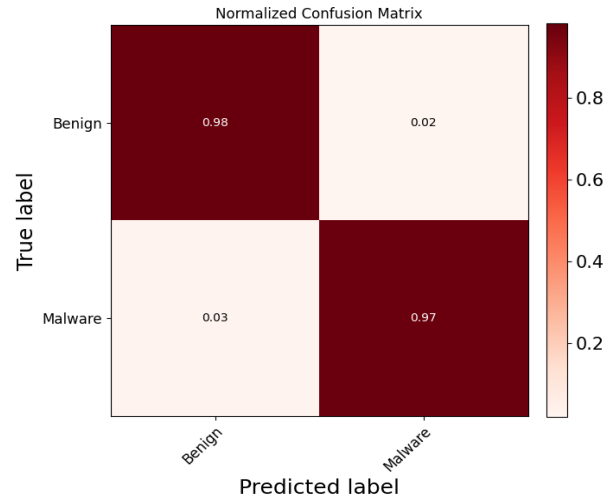


Fig. 4 Confusion Matrix

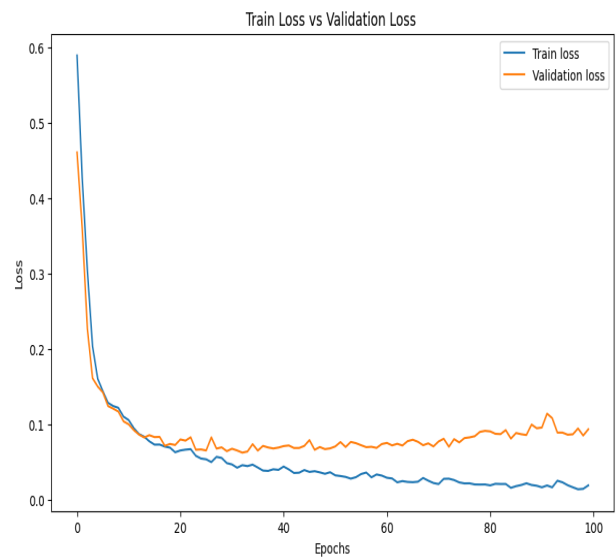


Fig. 5 Loss Plot

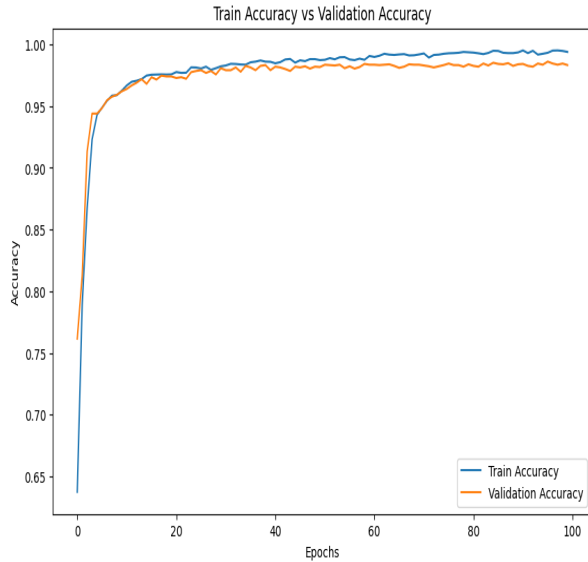


Fig. 6 Accuracy Plot

TABLE I. EVALUATION METRICS USED FOR MALWARE DETECTION

| Metrics        | Abbreviation | Definition                                 |
|----------------|--------------|--------------------------------------------|
| True Positive  | TU           | Instances correctly identified as malware  |
| True Negative  | TV           | Instances correctly identified as benign   |
| False Positive | FX           | Instances wrongly identified as malware    |
| False Negative | FY           | Instances wrongly identified as benign     |
| Precision      | PS#          | $TU/(TU+FX)$                               |
| Recall         | RE#          | $TU/(TU+FY)$                               |
| F1 Score       | FS#          | $2 \times PS# \times RE# / (PS# + RE#)$    |
| Accuracy       | ACC*         | Percentage of malware identified correctly |

TABLE II. NO. OF HYPERPARAMETER USED IN THE EXPERIMENT

| Model Parameters    | Values              |
|---------------------|---------------------|
| Activation function | ReLU, tanh, softmax |
| Loss Function       | Binary_crossentropy |
| Optimizer           | Adam                |
| Learning rate       | 0.001               |
| Batch size          | 500                 |
| Epochs              | 100                 |

TABLE III. RESULT COMPARISONS OF THE PROPOSED MODEL ON CICANDMAL2019 DATASET

| Method         | Year | PS# (%) | RE# (%) | FS# (%) | ACC* (%) |
|----------------|------|---------|---------|---------|----------|
| DeepAMD[27]    | 2021 | 93.50   | 93.40   | 93.20   | 93.40    |
| AMDI-Droid[28] | 2022 | 94.26   | 94.16   | 94.19   | 94.16    |
| SVM[29]        | 2022 | 95.9    | 82.6    | 88.8    | 94.36    |
| OEL-AMD[30]    | 2022 | 95.99   | 94.89   | 95.98   | 96.95    |
| DL-CNN[31]     | 2022 | 91      | 91      | 91      | 90.0     |

|                           |      |       |       |       |       |
|---------------------------|------|-------|-------|-------|-------|
| DL-LSTM[32]               | 2023 | -     | -     | 94.9  | 94.59 |
| MSerNetDroid[33]          | 2023 | 97.62 | 97.05 | 97.07 | 96.48 |
| Proposed XAI-AMD-DL Model |      | 97.75 | 97.76 | 97.75 | 97.98 |

## V. CONCLUSION

Increasing growth in android based smart devices and applications are mostly targetable by malware attackers in order to steal the valuable information and infect the system files. Therefore, this paper, proposed an XAI based hybrid CNN and Bi-GRU Android Malware Detection (AMD) framework using DL models. The proposed model is evaluated the using the CICAndMal2019 android malware dataset. The results obtained by the proposed XAI-AMD-DL model is 97.98% accuracy, and 97.75 %, 97.76%, 97.75% precision, recall and f1score, respectively. The aforementioned results of the model shows that the presented method is effective for classification of malware and benign samples. The future scope is to extend the work for multiclass malware detection by using latest metaheuristic algorithm for optimal feature selection and further improve the detection accuracy of the proposed model.

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